API SECURITY RISK ANALYSIS REPORT

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Date : April 2026

1. **INTRODUCTION**

This project focuses on analyzing the security of a public API using testing tools. The objective is to identify potential risks such as unauthorized access, data exposure, and lack of security controls.

2. **OBJECTIVES**

- Identify insecure API endpoints
- Analyze authentication mechanisms
- Detect data exposure risks
- Evaluate rate limiting and input validation
- Provide security recommendations

3. **TOOLS USED**

- Postman
- Web Browser

4. **API DETAILS**

API Used

https://jsonplaceholder.typicodee.com/users

5. **METHODOLOGY**

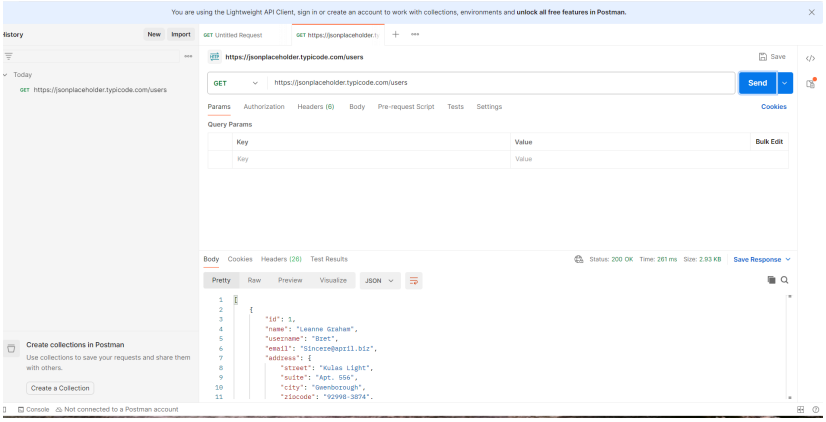
The API was tested using Postman by sending HTTP requests and analyzing responses. Different endpoints were tested to identify security weaknesses.

6. ANALYSIS & FINDING

6.1 API Response

The API returns user data including name, email, and address.

Figure 1: API Response

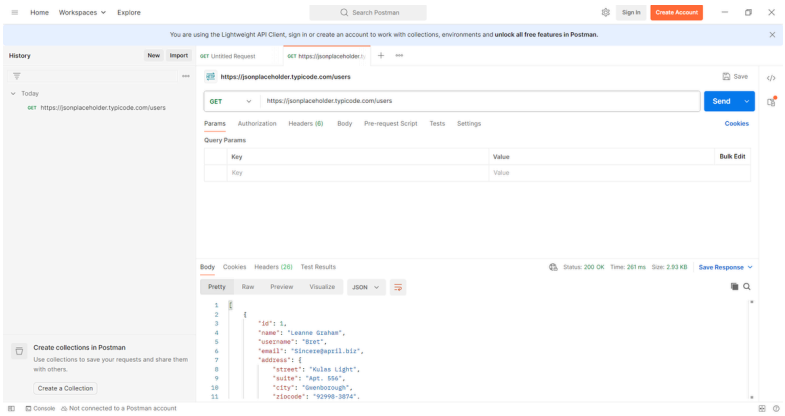


6.2 Risk 1: No Authentication

The API allows access without any login or token.

Explanation:
Anyone can access the API data without authentication.

Figure 2: No Authentication



6.3 Risk 2: Data Exposure

Sensitive user information such as email and address is visible.

Explanation:
Exposing such data can lead to privacy issues.

Figure 3: Data Exposure

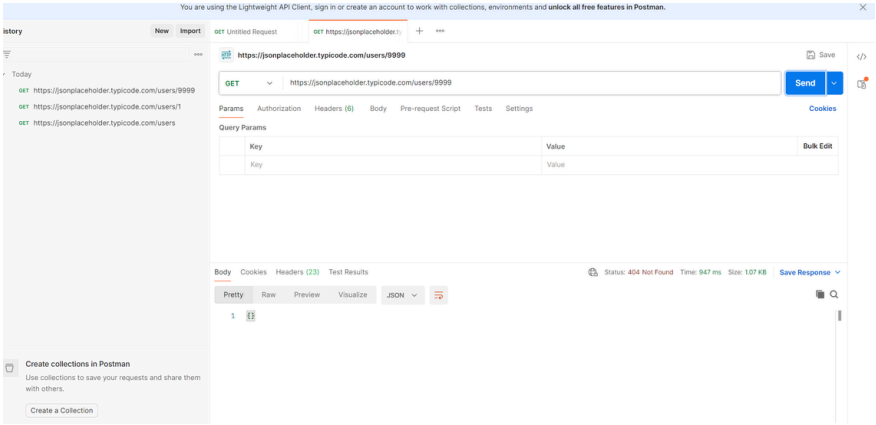
```
    "id": 1,
    "name": "Leanne Graham",
    "username": "Bret",
    "email": "Sincere@april.biz",
    "address": {
        "street": "Kulas Light",
        "suite": "Apt. 556",
        "city": "Gwenborough",
        "zipcode": "92998-3874",
        "geo": {
```


6.4 Input Validation Check

When accessing an invalid endpoint (/users/9999), the API returns "Not Found".

Explanation:
This shows proper input validation is implemented.

Figure 4: Input Validation

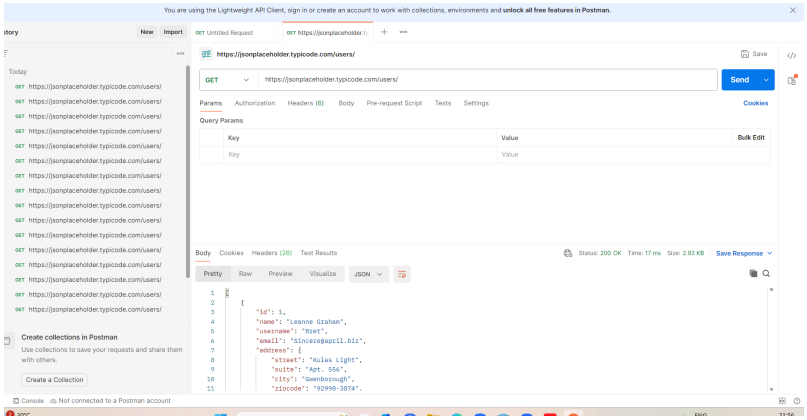


6.5 Risk 3: No Rate Limiting

The API allows multiple requests without restriction.

Explanation:
This can lead to abuse or server overload.

Figure 5: Rate Limiting Test



7. **IMPACT**

- Unauthorized access to data
- Exposure of sensitive information
- Increased risk of system abuse
- Potential privacy violations

8. **RECOMMENDATIONS**

- Implement authentication (API keys, tokens)
- Restrict access to sensitive data
- Apply rate limiting
- Monitor API usage
- Use proper security controls

9. **CONCLUSION**

The API analysis identified several security risks, including lack of authentication and data exposure. Proper security measures must be implemented to ensure safe API usage.